

Feature engineering

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graph TD; FE[Feature engineering] --> FT[Feature transformation]; FE --> FC[Feature construction]; FE --> FS[Feature selection]; FE --> FE2[Feature extraction]; FT --- FT_desc[change a column's form]; FT --- FT_list[missing values, categorical data, outliers, feature scaling, binning, log, ...]; FC --- FC_desc[build new columns by hand]; FC --- FC_list[combine columns, split a column, group values]; FS --- FS_desc[keep the useful columns]; FS --- FS_list[forward selection, backward elimination]; FE2 --- FE2_desc[build new columns by an algorithm]; FE2 --- FE2_list[PCA, LDA, t-SNE]; FS_list --- BR[both reduce the number of columns]; FE2_list --- BR;
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Feature transformation

change a column's form

missing values
categorical data
outliers
feature scaling
binning, log, ...

Feature construction

build new columns by hand

combine columns
split a column
group values

Feature selection

keep the useful columns

forward selection
backward elimination

Feature extraction

build new columns by an algorithm

PCA
LDA
t-SNE

both reduce the number of columns